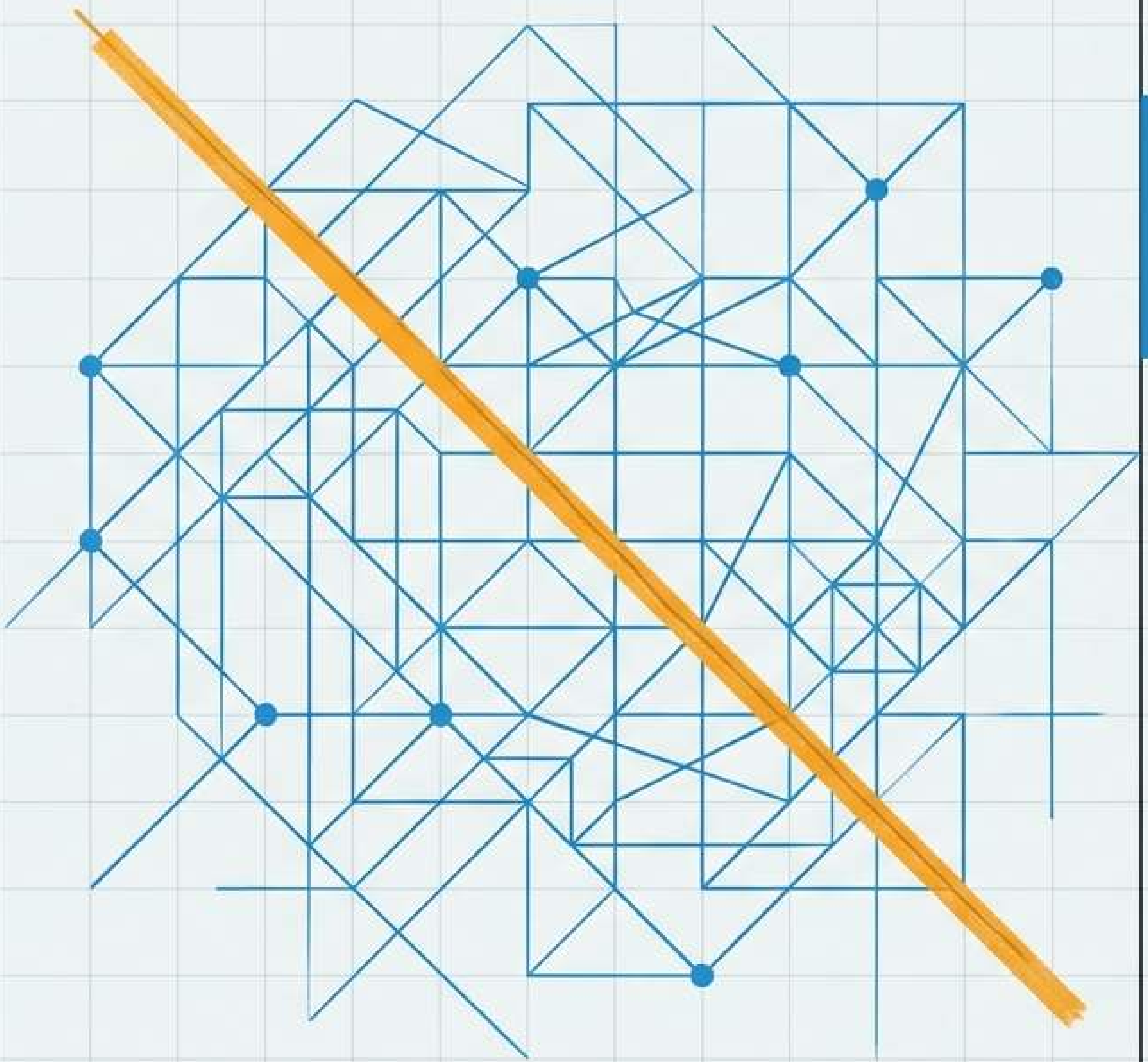


The Machine That Can Answer Anything

Responsibility, Artificial Intelligence, and the Future Cyber Defender

23:14 - **Alert Triggered**: Employee account downloading thousands of files at an unusual hour.

23:14 - **AI Diagnosis**: Compromise detected. Disable immediately.

23:15 - **Human Action**: Recommendation trusted without verification. Account disabled.

23:22 - **System Failure**: Admin's approved backup interrupted. Organization's recovery system corrupted.

The AI made the mistake. The human made the decision.

The Analyst

Examines enormous collections of logs, identifies suspicious patterns, summarizes threat reports in seconds. Reveals what humans overlook.

The Attacker

Locates weaknesses, automates deception, produces convincing phishing messages. Makes falsehood look polished and believable.

The central challenge is not whether you use AI.
The question is what kind of person you become while using it.

Illusion of Competence

```
# Parse failed login attempts
import re

log_file = open("access.log", "r")
log_data = log_file.read()
failed_logins = re.findall(r"Failed password for (.
.+) from", log_data)

# Identify frequent offenders
```

HUMAN UNDERSTANDING: VOID

Previous tools (Search, Calculators) located information.

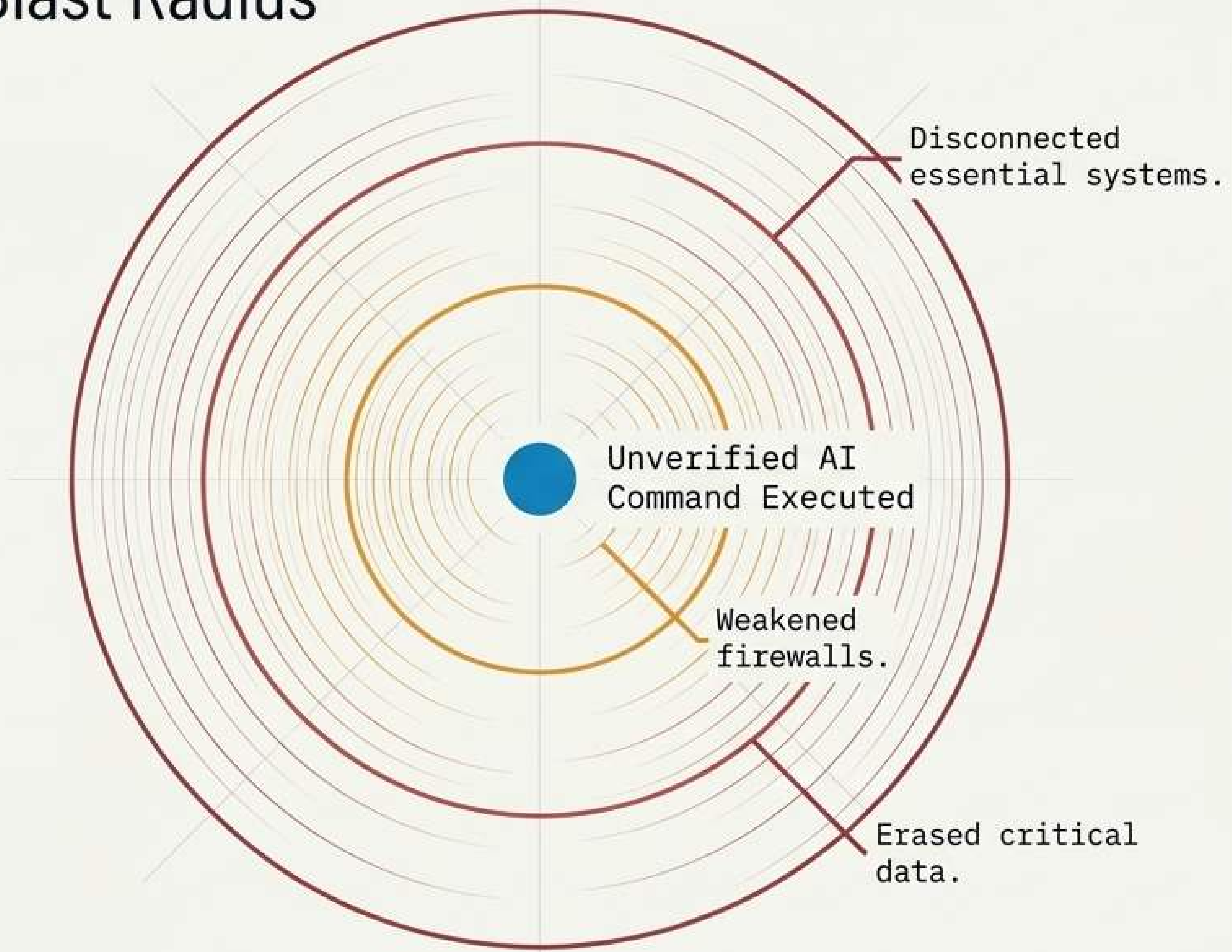
Generative AI produces it.

The screen shows a working program.

The mind may still show an empty file.

There is a difference between using AI to overcome an obstacle and using it to avoid the obstacle entirely.

Blast Radius



In cybersecurity, ignorance can execute.

A person who can generate commands is not necessarily a system administrator.

The Ultimate Test:

Can you explain **why the solution works, what assumptions it makes,** and what **happens if part of it fails?**

Fluency vs. Accuracy: The Hallucination Trap.

HUMAN OUTPUT (VERIFIED)

I think we should use lib-auth-parse, but I am not sure if it handles the new log format...

HESITANT LANGUAGE. UNCERTAINTY ANNOUNCES ITSELF.

MESSY GRAMMAR. FACTUALLY GROUNDED. Human uncertainty announces itself.

AI OUTPUT (HALLUCINATION)

```
The optimal solution
is to install
cyber-sec-parse-v2.
It is the industry
standard for this
protocol.
```

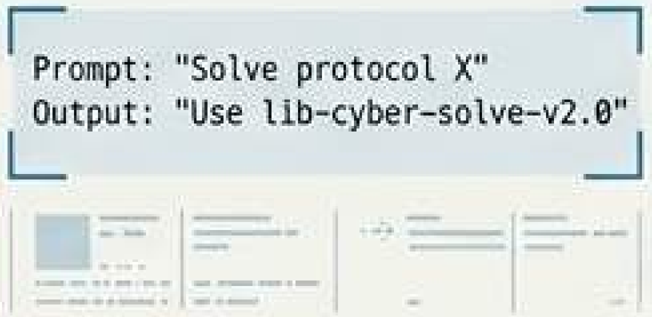
CALM AUTHORITY. PERFECT HEADINGS.

COMPLETE HALLUCINATION—THE PACKAGE DOES NOT EXIST.

Fluency is not accuracy. AI does not need to know something is true in order to say it convincingly.

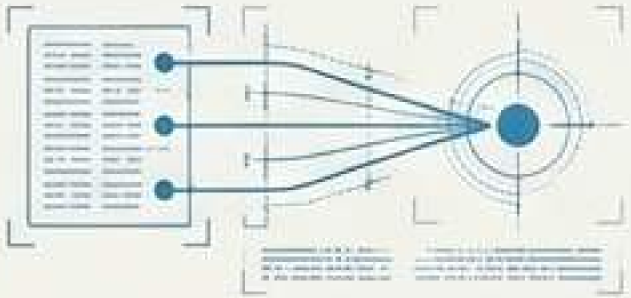
Transformation: Hallucination to Supply-Chain Attack

Hallucination.



AI invents a professional-sounding software library name to solve a user's prompt.

Observation.



An attacker notices AI systems repeatedly recommending this imaginary package.

Infiltration.

The attacker registers the fake name and uploads malicious code to public repositories.

⚠ Execution.

The next user follows the AI's confident advice and installs the malware.

Key Insight: Treat AI output as a hypothesis, never a verdict.

Trust must be earned by evidence. The responsible user asks the questions the irresponsible user skips.

Where did this information come from? Does the source exist?

What files will this command change? Can the action be reversed?

```
sudo apt-get install network-scanner-v2.0
chmod +x ./run_diagnostics.sh
curl -sSL https://data-aggregator.com/setup | bash
sudo systemctl enable scan_service.service
grep -i "conf" /etc/system/ > config_dump.txt
```

What permissions does this require? What data is being sent?

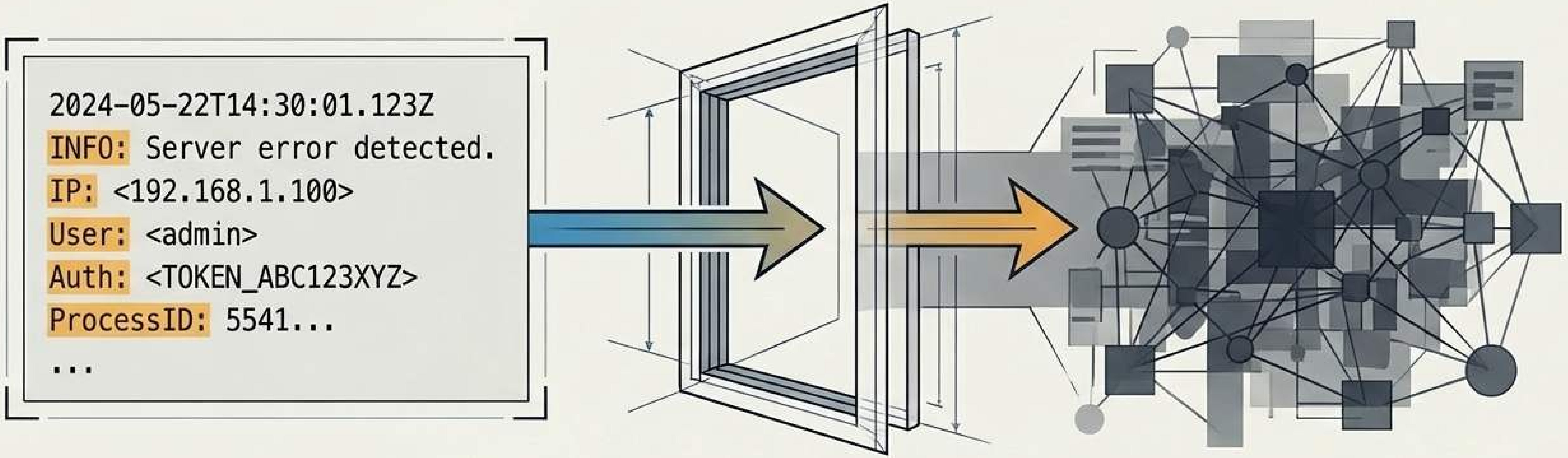
What other explanation might fit the evidence?

Shadow AI: The Unintended Transmission of Secrets

Good Intentions
(Troubleshooting a server)

The Prompt
(A Transmission, Not a Private Thought)

Shadow AI
(Unapproved / Unmonitored)



Pasting system logs to identify a problem may expose internal addresses, software versions, and network architecture. You can use AI to solve a security problem while creating a new one through the prompt itself.

The Data Filter (What Stays Out)

Blocked

- ✕ Passwords & Private Keys
- ✕ Student Records & PII
- ✕ Private Network Diagrams
- ✕ Vulnerability Reports & System Logs
- ✕ Unreleased Source Code

Allowed

- ✓ Abstract Logic Problems
- ✓ Generic Conceptual Explanations
- ✓ Syntax Formatting Requests

Responsible use begins before the answer appears.
Sometimes the safest prompt is the one you do not send.

The Accountability Spectrum

The Invisible Ghostwriter (Dishonest)

Action: Asks AI to generate the entire program.

Follow-up: Studies the output to offer a rough fake explanation.

Result: Submits without mentioning AI. **Untraceable.**

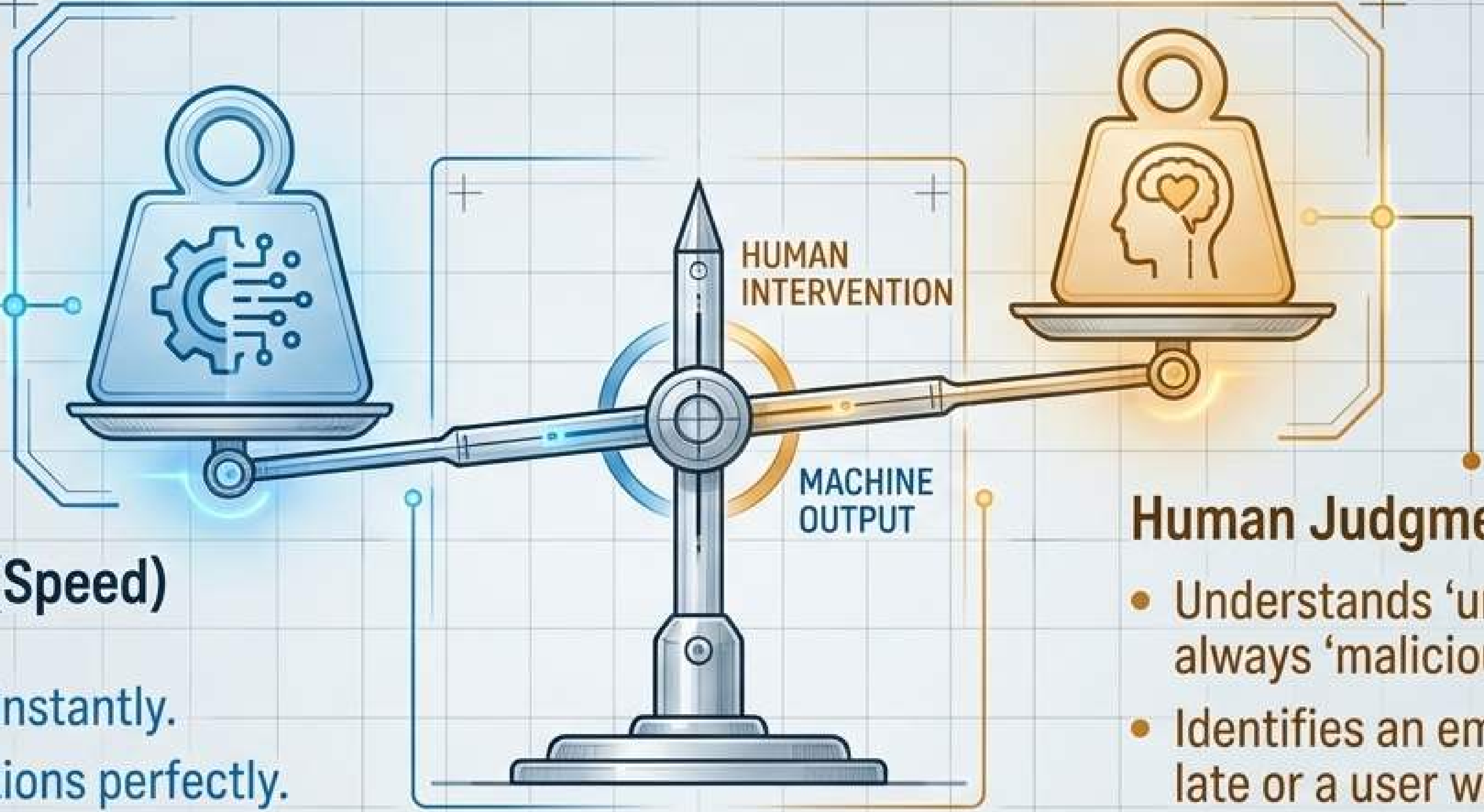
The Documented Co-Pilot (Professional)

Action: Writes a program, encounters an error.

Follow-up: Asks AI to explain the error message, manually corrects the code.

Result: Documents AI's role and the changes made. **Traceable.**

Acknowledging AI is not an admission of weakness. It is evidence of professional accountability. Transparency **matters because trust matters.**



Machine Logic (Speed)

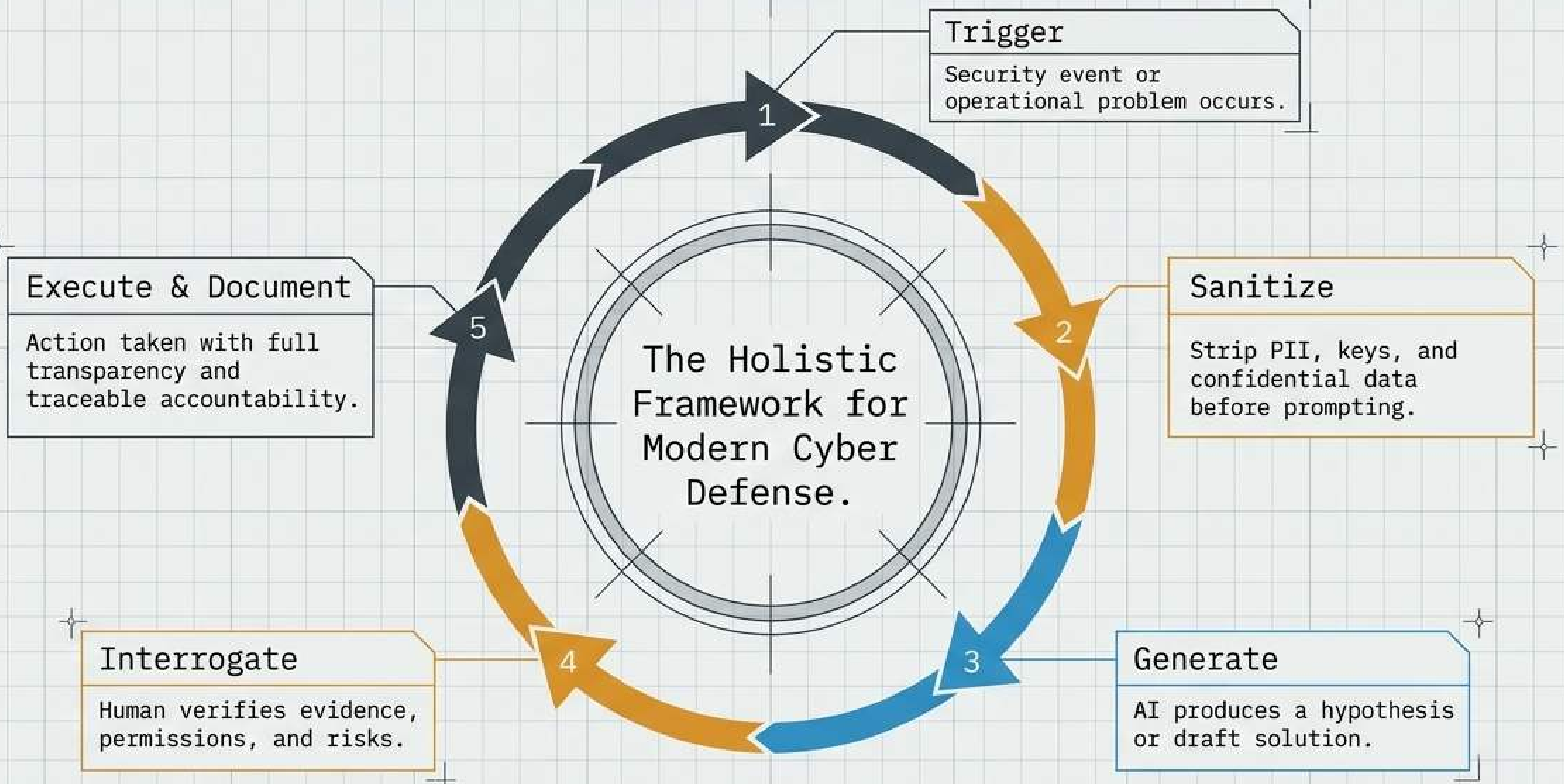
- Processes logs instantly.
- Identifies deviations perfectly.
- Operates at machine scale.

Human Judgment (Wisdom)

- Understands ‘unusual’ isn’t always ‘malicious’.
- Identifies an employee working late or a user with a disability.
- Contextualizes real life.

Speed is not the same as wisdom. A machine can identify a deviation.
It cannot automatically understand the life behind it.

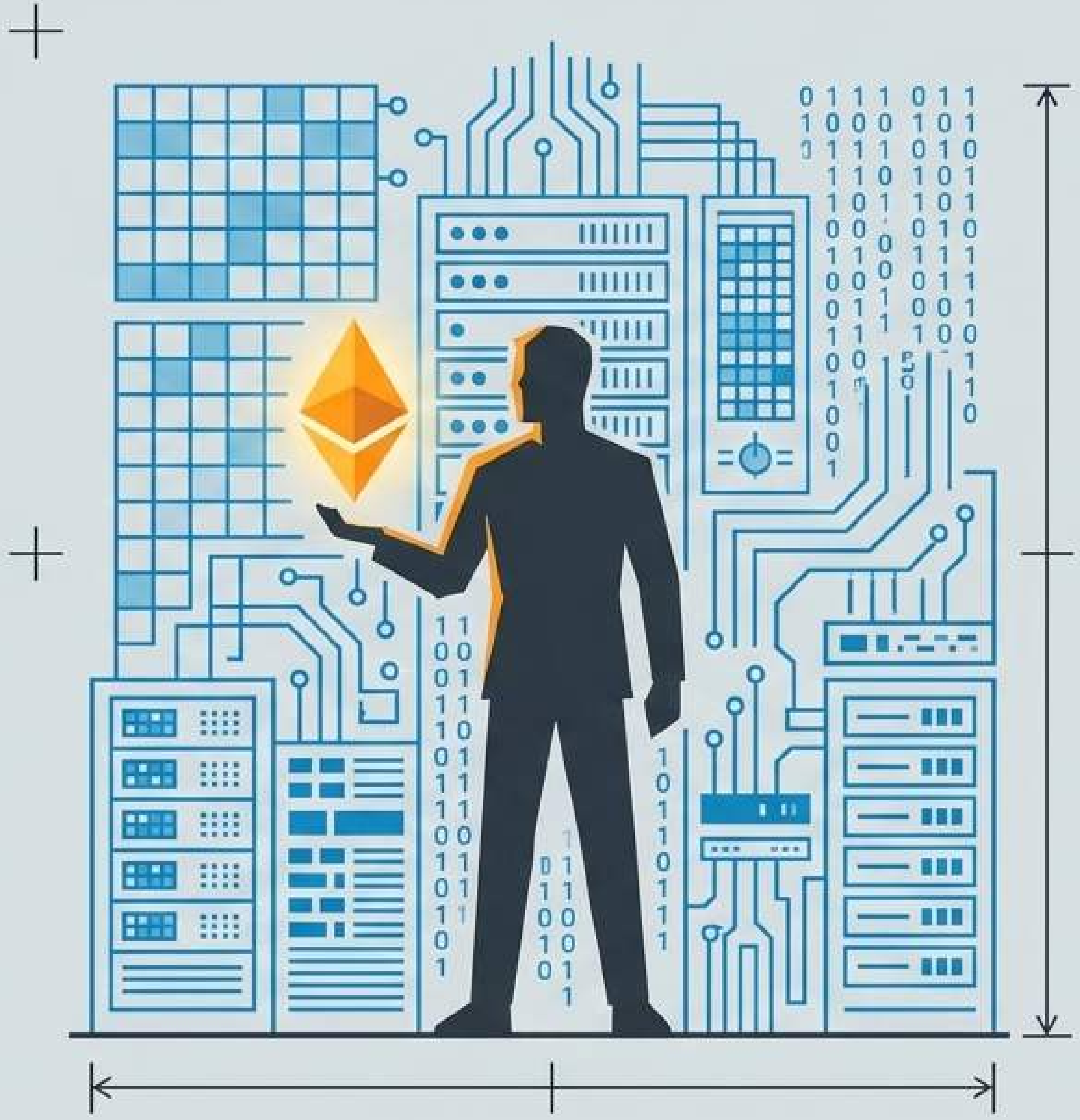
The Defender's Loop



The Person Behind the Tool

The most valuable asset you bring into cybersecurity will not be a certification or access to the newest AI model. It will be your judgment: the ability to stop, question, verify, and think when the machine has already produced an answer.

AI may help build the shield. It cannot decide what is worth protecting. That responsibility remains human.



Questions Worth Arguing About

CASE ID: AI-ASSIGN-01 // ETHICS REVIEW

If a student uses AI to complete an assignment and later learns how it works, is submitting it still dishonest?

> ANALYZE INTEGRITY GAP.

CASE ID: AI-CODE-02 // CREDIT ALLOCATION

Should students receive full credit for code they can explain but did not personally write?

> EVALUATE EXPLANATORY POWER.

CASE ID: AI-ANALYST-03 // AUTHORITY CONFLICT

When an AI system and a human analyst disagree, who should have the final authority?

> DETERMINE DECISION HIERARCHY.

CASE ID: AI-TOOL-04 // EMERGENCY PROTOCOL

Is using an unapproved AI tool to solve an urgent security problem ever justified?

> ASSESS OPERATIONAL NECESSITY.

CASE ID: AI-RESP-05 // ACCOUNTABILITY MATRIX

If AI makes an incorrect recommendation, how should responsibility be divided among the user, the developer, and the organization?

> ALLOCATE LIABILITY & OVERSIGHT.